

EVAN KERR

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Education

McMaster University

September 2021 – December 2026

Bachelor of Engineering in Mechanical Engineering

Hamilton, ON

- **cGPA:** 3.85/4.0 (10.7/12.0)
- **Awards:** Senate Scholar - 2025/26 (Top 1% Fall-Winter Average), Dean's Honours List (x4)
- **Specialization:** Smart Systems (Sensing & Instrumentation, Robotics, Control Systems, Manufacturing Processes)

Technical Skills

Design & Analysis: Solid Edge, Inventor, FEA (ANSYS, NX Nastran), AutoCAD, GD&T, DFMA, Teamcenter PLM

Manufacturing: Machining (Lathe, Milling, Drill Press), Welding, Rapid Prototyping, RCA, DOE

Languages/Tools: Python, MATLAB, Arduino, Simulink, Excel & MS Office, MathCAD, PowerBI, Bluebeam

Experience

Hatch

May 2024 – August 2025

Mechanical Engineering Intern

Mississauga, ON, In-Person

- **Designed & tested** sensor fixture equipment for **Machine Vision** applications in **Autodesk Inventor**, maintaining calibration resolution of **50nm**. Verified **DFM** compliance for **10+** sheet metal parts with vendors.
- Verified **thermal expansion** compliance for **furnace heat shields** using tolerance stack-up analysis, ensuring designed **hydraulic** equipment service life under high-temperature conditions. Drafted package **fabrication & general assembly drawings** reviewed in **Bluebeam**.
- Developed a **Python**-based probabilistic **corrosion model** based on **NDT** inspections, streamlining data entry to reduce setup time by **74%**. Validated statistical accuracy in **MathCAD**.
- Drafted multi-phase **steel plant layouts** in **AutoCAD** for a process feasibility study. Integrated design feedback through weekly **design reviews**, as well as weeklong workshops on site with **30+ international stakeholders**.
- Contributed to the development of **steel fabrication standard specifications**, saving **\$1k** in project costs per year.

Ericsson

May 2023 – August 2023

Project Planner Intern

Mississauga, ON, Hybrid

- Optimized **Project Controls** workflows for **KPI** dashboards & **QA** documentation, directly supporting Engineering Managers in tracking deliverables and maintaining **audit-readiness**.
- Managed data organization for **70+** ongoing project plans under **20+** PMs using **Microsoft Project** and **Excel**, reducing reporting errors by **95%**.

Projects

Conforming Pneumatic End Effector

- Collaborated with **Honda** to design an adaptive pneumatic end effector for automotive assembly, achieving a **10kg** payload capacity and a **75mm** grip stroke for varied component geometries.
- Re-engineered assembly equipment using **ANSYS & Inventor** to validate structural safety factors, increasing lifting capacity by **80%** while eliminating critical **failure modes**.
- Developed a **spring sizing calculator** in **Excel** designed for worst-case loading configurations, reducing spring selection time by **83%**.

Extracurriculars

FUSE, McMaster University

September 2025 – Present

Teaching Assistant - Engineering Design

McMaster University

- Assisting **40+** students in weekly design labs, applying **Autodesk Inventor** to develop students' **2D/3D CAD** skills.
- Preparing lab material & grading assignments on a **weekly** basis, **reinforcing topics** in engineering design and providing **tangible feedback** to improve technical skills.

McMaster Society of Mechanical Engineers

September 2023 - Present

Year Representative

McMaster University

- Representing **800+** mechanical engineering students in bi-weekly meetings, communicating feedback to faculty and contributing to measurable course delivery improvements.

MAC Formula Electric

September 2022 - September 2023

Manufacturing Composites Subteam Member

McMaster University

- Assisted in **CFRP** panel **fabrication & testing** for the vehicle chassis, following SAE CFRP installation standards.